SECTION 5 – POWER REQUIREMENTS

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5–1 INTRODUCTION

Signa 1.5T TwinSpeed requires the Power Distribution module (PD1) located in the lower portion of the ACGD/PDU Cabinet. Prior PDU configurations are not compatible with the Signa 1.5T TwinSpeed.

For systems with ACGD/PDU Cabinet installed the existing PDU will be used. For systems without ACGD/PDU Cabinet the ACGD Upgrade will replace the existing system PDU. The ACGD upgrade includes the new Power Distribution module (PD1) in the lower portion of the ACGD/PDU Cabinet (MR3) which distributes power to most MR system components and provide the proper voltage configuration for the ACGD modules. Refer to Table 5–1 for the required customer power specifics. Refer to Section 5–2 CRITICAL POWER REQUIREMENTS for specifications of required facility input for the Signa 1.5T TwinSpeed system.

WARNING!

THE FACILITY TRANSFORMER AND FEEDER WIRES NEED TO BE CORRECTLY SIZED FOR THE SIGNA SYSTEM WITH ACGD.

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

Customers should carefully consider the advantages and disadvantages of raised flooring, access flooring, conduits, floor ducts, and surface raceways for running cables in accordance with local and national codes. If used, conduits should be large enough to pass any cable and its connector through with all other cables in the conduit.

To reduce voltage regulation problems and wiring costs, minimize the cable length between the primary power source and the Power Distribution Unit. When routing cables, keep all phase conductors and ground for a circuit in the same trough. Whenever possible, keep power cables away from signal and data cables. Use separate trough or dividers in duct.

5–1 INTRODUCTION (Continued)

TABLE 5–1
REQUIRED CUSTOMER POWER

MR COMPONENT	VOLTAGE (VAC)	FREQUENCY	PHASE	MAX. AMPS	COMMENTS
GE Main Disconnect Panel (MDP) See Note1, 2 & 3 See Note 4 if customer MDP	480Y/277 VAC ±10% or 400Y/230 VAC ±10%	60 Hz 50 Hz	(3+GND) See Com- ments	See Note 5	Recommend input configuration: 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to PDU or inside the Main Disconnect Panel and not brought to the Power Cabinet. (See Note 6) Optional input configuration: 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.
Magnet Rundown Unit	100–120 or 200–240	50/60 Hz	1	1.0	Hard wired in unit.
Pneumatic Patient Alert Control Box	115 or 230	50/60 Hz	1	0.2	An outlet on the Operator Workspace Table equipment can be used.
Service Receptacle in Magnet Room	110–120 See Comments	50/60 Hz	1	2.0	Receptacle required for small power tools. Local voltage and portable transformers for voltages values.
*O_2 Monitor	110–120 or 200–240	50/60 Hz	1	3.0	Hard wired in monitor
*MRI Music System	120	60 Hz	1	1.2	Receptacle required in Control Room
*Advantage Windows	Refer to Direction *2261309–100 Advantage Workstation Ultra60 Pre–Installation.*				

Note * Optional equipment.

1 Power phase conductors, neutral (if present), and ground conductor must be routed inside the same raceway, cable tray, trench cable. or cord per National Electric Code (NEC) 2002 Articles 250.134, 300.3, 517.13 or 1999 Articles 250–134, 300–3, 517–13.

2 Signa TwinSpeed MDP controls power to the following system equipment:
– Power Distribution Unit
– System Cooling Cabinet
– Shield/Cryo Cooler Compressor
– Magnet Monitor equipment including the Magnet Monitor, Modem, Uninterruptible Power Supply (UPS*) (optional) for Magnet Monitor, Multiplexer Box (optional).

3 MDP power circuits for Twin System Cooling Cabinet and Shield/Cryo Cooler Compressor Cabinet, and cooling for these units, are required immediately upon magnet arrival. If permanent site power is not ready, temporary power drop line and cooling must be made available. If site voltage is not any of the voltages listed above, customer must provide transformer and secondary circuit breaker to provide correct voltage and/or configuration. Refer to Section 1–5 **FACILITY OPTIONS** for listing of step up transformers options.

4 If customer provided MDP has been selected based on meeting **SECTION 1 – SYSTEM CONFIGURATION** upgrade flowcharts requisites then customer provided MDP MUST meet all MDP requirements.

5 Maximum amps dependent on voltage selected. Refer to Section 5–2, CRITICAL POWER REQUIREMENTS, for configuration.

6 PDU Module is located in the lower portion of the ACGD/PDU Cabinet (MR3).

5–2 CRITICAL POWER REQUIREMENTS

The system Main Disconnect Panel is a Low Voltage Low Energy local control unit, with multi–point remote control capability, connected to the feeder lines that supply input power to the following:

- Power Distribution Unit (PD1)
- Shield/Cryo Cooler Compressor Cabinet
- System cooling equipment: Twin System Cooling Cabinet (TSCC) or Twin Gradient Water Cooler (TGWC)
- Magnet Monitor equipment.

The Main Disconnect Panel is provided as part of the system per **SECTION 1 – SYSTEM CONFIGURATION** flowcharts for upgrades system cooling equipment configurations options and relationship to other system equipment/options selections.

WARNING!

IF CUSTOMER PROVIDED MDP HAS BEEN SELECTED BASED ON MEETING THE REQUISITES IN SECTION 1 – SYSTEM CONFIGURATION UPGRADE FLOWCHARTS THEN CUSTOMER PROVIDED MDP MUST MEET ALL MDP REQUIREMENTS.

All work is to be done in accordance with national and local electrical codes. Refer to Section 5–3, POWER DISTRIBUTION SYSTEM, for Main Disconnect Panel capability and set up.

WARNING!

THE FACILITY TRANSFORMER AND FEEDER WIRES NEED TO BE CORRECTLY SIZED FOR THE SIGNA SYSTEM WITH ACGD.

IF AN UNINTERRUPTIBLE POWER SUPPLY (UPS) WILL BE PROVIDING POWER TO THE ENTIRE MR SYSTEM THEN THERE IS A NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS.

5–2 CRITICAL POWER REQUIREMENTS (Continued)

Configuration:

- Recommend input configuration 3 phase Grounded WYE with Neutral and Ground (5 wire system). Note, Neutral must be terminated prior to or inside the Main Disconnect Panel and not brought to the ACGD/PDU Cabinet.
- Optional input configuration 3 phase DELTA with Ground (4 wire) input, recommend corner Grounded Delta configuration.

Frequency: 50 ± 3 Hz or 60 ± 3 Hz

Regulation: 4% maximum at system maximum 5 seconds power demand from source to PDU (i.e. includes all feeders and transformers to utility).

Phase Balance: Difference between the highest phase line–to–line voltage and the lowest phase line–to–line voltage must not exceed 2%.

Daily Voltage Variation: ±10% from nominal under worst case line and load regulation.

PDU Voltage: 200, 400, 480 VAC ±10%

Shield/Cryo Cooler Compressor Voltage: 380/400/415 VAC 50 Hz or 460/480 VAC 60Hz

TSCC or TGWC (excludes Water Cooled TGWC powered from PDU) Voltage: 400 VAC ±10% 50 Hz or 480 VAC ±10% 60Hz

Magnet Monitor equipment Voltage: 100/120 or 200/240 VAC 50/60 Hz

Voltage Transients: Phase–to–phase voltages must be within 2% of the lowest phase–to–phase voltage. Maximum allowable transient voltage above or below nominal waveshape not to exceed 200 V at a maximum duration of 1 cycle and frequency of 10 times per hour.

Facility Zero Voltage Reference Ground:

- Main facility ground wire to be minimum 1/0 AWG copper or same size as power feeder wire, which ever is larger.
- Main facility ground wire to be insulated.
- Ground impedance to earth at power source to be 2 ohms or less.
- Main facility ground wire to be bonded at every distribution box in an approved grounding block.

5–2 CRITICAL POWER REQUIREMENTS (Continued)

Maximum Momentary Demand: The power demands for the Signa 1.5T Twin*Speed* system with ACGD are specified as a function of the duration of the power demand. Table 5–2 lists points on the curve.

The power system feeding the Signa system with ACGD must be designed to meet the specifications of less than 4% regulation when loaded at 92.4 KVA (corresponds to 5.0 second allowable consumption). For short intervals the Signa system with ACGD power demands can exceed 56.2 KVA and the line voltage delivered to the system will sag below the 4% regulation. The Signa system with ACGD is designed to tolerate these **short** voltage sags.

TABLE 5–2
SYSTEM PEAK POWER DEMAND

COOLING EQUIPMENT	**TSCC *See Note 1***			**TGWC *See Note 2***		
SYSTEM EQUIPMENT	**OUTDOOR TSCC**	**INDOOR WATER COOLED TSCC**	**INDOOR AIR COOLED TSCC**	**OUTDOOR TGWC**	**INDOOR WATER COOLED TGWC**	**INDOOR AIR COOLED TGWC**
PDU draw for 5.0 sec	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA	~56.2 KVA
PDU draw for 1.0 sec	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
PDU draw for 0.1 sec	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
PDU draw for 0.04 sec ***See Note 3***	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA	~65 KVA
Magnet Monitor ***See Note 4***	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	23 KVA	20 KVA	22.7 KVA	16 KVA	0 KVA ***See Note 5***	11.6 KVA
TOTAL for 5.0 sec	~92.7 KVA	~89.7 KVA	~92.4 KVA	~85.7 KVA	~69.7 KVA	~81.3 KVA
TOTAL for 1.0 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA
TOTAL for 0.1 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA
TOTAL for 0.04 sec	~101.5 KVA	~98.5 KVA	~101.2 KVA	~94.5 KVA	~78.5 KVA	~90.1 KVA

Note
1 TSCC provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil.

2 TGWC provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor. Customer provided water cooling equipment power demands are not included in the TGWC values.

3 The PDU draw on the line will not exceed list values. The ACGD Power Supply may provide up to 170 KVA for 0.003 seconds from supply internal capacitance but the supply will recharge capacitors at a power level less than 65 KVA.

4 The Magnet Monitor equipment power is 1.5 KVA 1 phase on an unbalanced leg of 3 phase input (4.5 KVA 3 phase equivalent).

5 The Indoor Water Cooled TGWC is powered from the PDU and therefore included in the PDU draw value.

5–2 CRITICAL POWER REQUIREMENTS (Continued)

Average (while scanning) Power Demand: Refer to Table 5–3

TABLE 5–3
SYSTEM AVERAGE (CONTINUOUS) SCANNING POWER DEMAND

COOLING EQUIPMENT	**TSCC** *See Note 1*			**TGWC** *See Note 2*		
SYSTEM EQUIPMENT	**OUTDOOR TSCC**	**INDOOR WATER COOLED TSCC**	**INDOOR AIR COOLED TSCC**	**OUTDOOR TGWC**	**INDOOR WATER COOLED TGWC**	**INDOOR AIR COOLED TGWC**
PDU draw *See Note 3*	43.1 KVA	43.1 KVA	43.1 KVA	43.1 KVA	44.5 KVA	43.1 KVA
Magnet Monitor	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA	4.5 KVA
Shield/Cryo Cooler Compressor	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA	9 KVA
System Cooling equipment (configuration indicated in column heading)	18.3 KVA	14 KVA	13.0 KVA	15 KVA	0 KVA *See Note 4*	8.8 KVA
TOTAL *See Note 5*	74.9 KVA	70.6 KVA	69.6 KVA	71.6 KVA	58.0 KVA	65.4 KVA

Note
1 TSCC Provides water cooling for Shield/Cryo Cooler Compressor and Gradient Coil.

2 TGWC Provides water cooling for Gradient Coil ONLY. Customer/site provided water cooling is required for Shield/Cryo Cooler Compressor. Customer provided water cooling equipment power demands are not included in the TGWC values.

3 The PDU is rated for 45 KVA continuous power.

4 The Water Cooled TGWC is powered from the PDU and therefore included in the PDU draw value.

5 GE Main Disconnect Panel (MDP) is rated continuous power draw is 77 KVA but MDP continuous draw does not exceed list demands.

Standby (no scan) Power Demand: 26.9 KVA at 0.9 lagging Power Factor including 4.4 KVA for PDU, 9 KVA for system cooling equipment, 9KVA (continuous operation) for Shield/Cryo Cooler Cabinet, and 1.5 KVA 1 phase for Magnet Monitor equipment (4.5 KVA 3 phase equivalent).

5–3 POWER DISTRIBUTION SYSTEM

5–3–1 Main Disconnect Panel (MDP)

WARNING!

If customer provided MDP has been selected based on meeting **SECTION 1 – SYSTEM CONFIGURATION** upgrade flowcharts requisites then customer provided MDP MUST meet all MDP requirements.

The Signa TwinSpeed system includes a Main Disconnect Panel (based on **SECTION 1 – SYSTEM CONFIGURATION** upgrade flowcharts) with multi–point remote control capability which is shown in Illustration 5–1. The Main Disconnect Panel consists of the following:

- A three–pole circuit breaker rated for the current of the PDU circuit. The short–circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A three–pole circuit breaker rated for the current of the System Cooling Cabinet circuit. The short–circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A three–pole circuit breaker rated for the current of the Shield/Cryo Cooler Compressor Cabinet circuit. The short–circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate facility available short circuit current.
- A circuit to provide 120VAC single phase power to the Magnet Monitor, Modem, UPS for Magnet Monitor (optional), and Multiplexer Box (optional). The short–circuit current interrupting rating of the breaker is 25,000 Amperes to accommodate available fault current. The MDP includes a single phase step down transformer for 120VAC loads such as Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box).
- The MDP Panel has receptacles inside the panel enclosure for connections of the UPS for Magnet Monitor input and output, Multiplexer Box, Magnet Monitor, and modem. The enclosure has provision for these cables to enter through the access panels in the bottom left side of the enclosure. Mounting of the panel must allow for 5–6 inch (127–152 mm) of free space to allow for cable bending and installation. Strain relief bushings are provided with the individual equipment for each of these cables, not provided with the MDP.

Main Disconnect Panel (MDP) is to be located so the top of the upper circuit breaker handle does not exceed 79 inches (2000 mm) from the floor when in the ON position and is visible to the customer and service personnel from the Power Distribution Unit (PD1), System Cooling Cabinet (TSCC or TGWC), Shield/Cryo Cooler Compressor Cabinet. The height of the MDP location is per National Electric Code (NEC) 2002 Article 404.8 or NEC 1999 Article 380–8 or the height specified in local/national governing code. The optional UPS for the Magnet Monitor may be located below the MDP if sufficient space is available or adjacent if sufficient space is not available, refer to Section 2–6–5 Magnet Monitor.

5–3–1 Main Disconnect Panel (MDP) (Continued)

Note

The MDP circuits for the system cooling equipment (TSCC or TGWC), the Shield/Cryo Cooler Compressor Cabinet, and the single phase transformer for Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box) have auto restart feature where normal power is returned after a time delay of 3 to 30 seconds (field adjustable) to reduce operating costs and minimize cryogen consumption of the system thus decreasing system downtime. The MDP Emergency Off circuit turns off power to all branch circuits including the Magnet Monitor UPS option output and turns off the auto restart function.

The PDU circuit has low voltage release feature which disconnects power from the PDU upon the first loss of power. Power to the PDU is not restored automatically after a power interruption. Emergency Off operation disconnects power from all circuits including the PDU.

For the system cooling equipment (TSCC or TGWC), the Shield/Cryo Cooler Compressor Cabinet, and Magnet Monitor equipment the auto restart function is disabled by the Emergency Off pushbuttons. Extended power outages of 2 days or more will result in the auto restart function loss of battery backup and will require pressing of the Main Power ON push button manual restart after restoration of normal power.

The circuit breakers or fuses ahead of the MDP must be capable of handling the magnetizing inrush currents of the System Cooling Cabinet, Shield/Cryo Cooler Compressor, Magnet Monitor equipment, and transformer of the PDU module (PD1) in the ACGD/PDU Cabinet (MR3). If fuses are used time delayed fuses are recommended.

Check local and national codes to determine if an interlock to the air–conditioning unit in the Computer/Equipment Room is required in the protective disconnect set–up.

The MDP provides two Emergency Off buttons to be connected to the MDP to disable the power to all system equipment in emergency situations. The Emergency Off buttons need to be mounted near each exit 60 in. (1524 mm) from the floor, near the exit in the Magnet Room and Equipment Room. The Emergency Off buttons are clearly labeled “Emergency Off” and visible to personnel. It is important that the buttons are labeled “OFF” and not “STOP” since there exists an “Emergency Stop” button in the Signa system which powers down only a portion of system equipment for patient safety.

Note

The emergency off circuit disconnects power to the PDU, Twin System Cooling Cabinet, Shield/Cryo Cooler Compressor Cabinet, and from the single phase transformer for Magnet Monitor equipment. Power can be restored to the MDP outputs by pressing the MAIN POWER ON pushbutton on the MDP for the TSCC, Shield/Cryo Cooler Compressor Cabinet, Magnet Monitor equipment (Magnet Monitor, optional UPS for Magnet Monitor, modem and the optional Multiplexer Box). Power to the PDU is restored by pressing the PDU POWER ON pushbutton and also requires pressing the EMO Reset button on the PDU.

5–3–1 Main Disconnect Panel (MDP) (Continued)

The Main Disconnect Panel (MDP) is lockable to meet OSHA requirements for power Lockout/Tagout requirements. The MDP provides for the disconnection of the facility power to the PDU, Twin System Cooling Cabinet, and Shield/Cryo Cooler Compressor Cabinet. Individual branch circuits for the PDU, Magnet Monitor equipment, Twin System Cooling Cabinet, and Shield/Cryo Cooler Compressor Cabinet consisting of lockable GE Spectra circuit breakers. The MDP also has electrical contacts for an interlock to the air–conditioning units in the Equipment Room. Check local and national codes to determine if an interlock to the air–conditioning unit in the Computer/Equipment Room is required in the protective disconnect set–up.

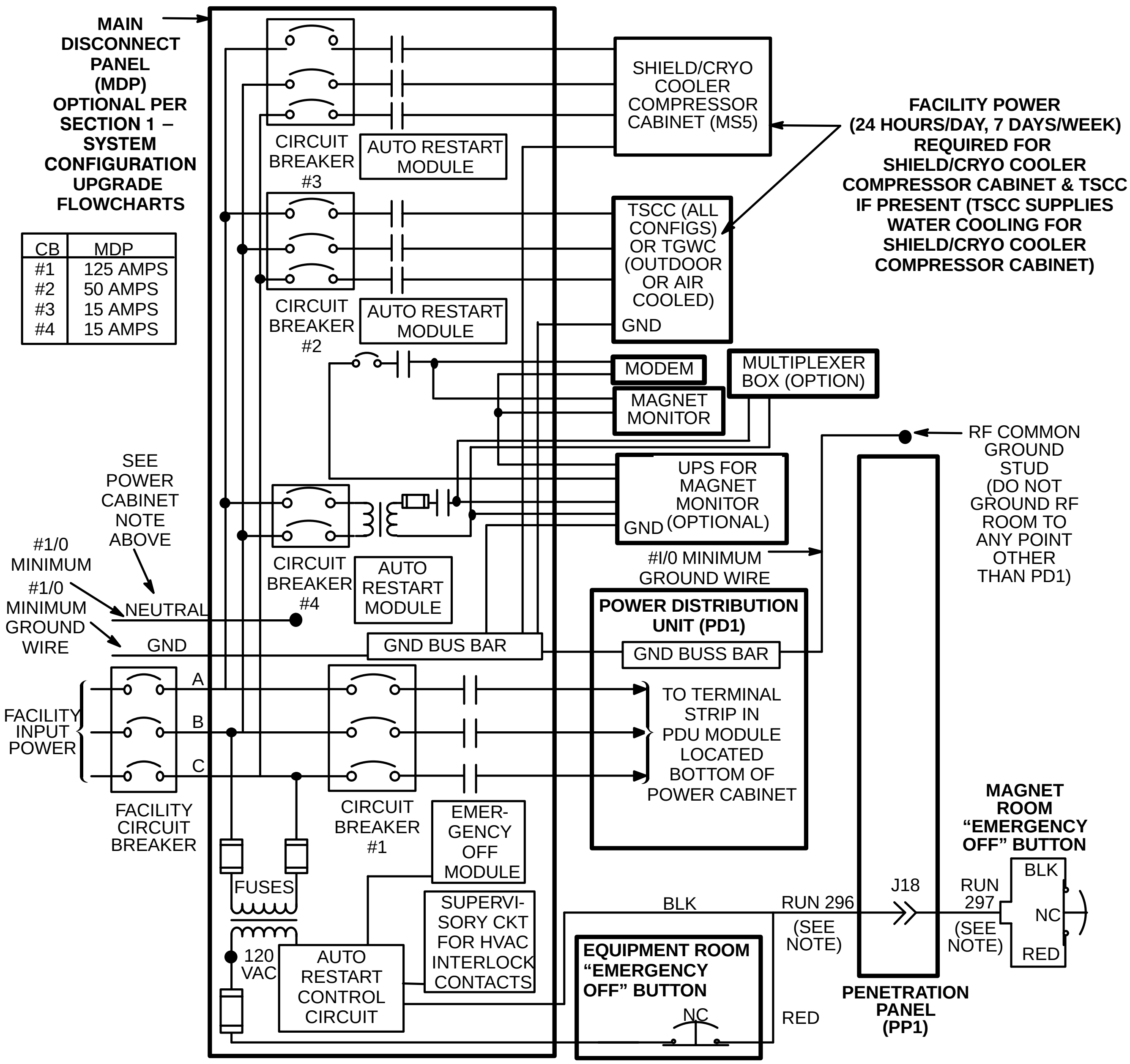
The GE available Main Disconnect Panel is UL labeled in accordance with 2002 National Electric Code (NEC) Article 110.2 or 1999 NEC Article 110–2. If the GE MDP is not utilized then the customer supplied MDP must be listed and labeled by a Nationally Recognized Testing Lab (NRTL) such as Underwriters Laboratory (UL) in accordance with 2002 National Electric Code (NEC) Article 110.2 or 1999 NEC Article 110–2.

Note

The maximum conductor the MDP can accept is #3/0 AWG (83 mm^2). For feeders larger than 3/0 AWG (83 mm^2) the wires must be reduced (ie. splice, junction box, etc.) to 3/0 AWG (83 mm^2) within 10 feet (3 meters) of MDP. It is important to note the maximum cable wire from the MDP to the PDU must not be larger than 2/0 AWG (70 mm^2).

5–3–1 Main Disconnect Panel (MDP) (Continued)

NOTE:
- RUNS 296 AND 297, & POWER CORDS FOR SCC & MAGNET MONITOR EQUIPMENT (MAGNET MONITOR, UPS INPUT & OUTPUT, MODEM, OPTIONAL MULTIPLEXER) ARE GE SUPPLIED CABLES. **ALL OTHER WIRING IS CUSTOMER SUPPLIED.**
- TWO REMOTE EMERGENCY “OFF” BUTTONS ARE SUPPLIED WITH SYSTEM MDP.
- CIRCUIT BREAKERS ARE PROVIDED FOR PDU, TSCC, SHIELD/CRYO COOLER COMPRESSOR CABINET, MAGNET MONITOR EQUIPMENT CIRCUITS.
- ALL BRANCH CIRCUITS DROP OUT ON LOSS OF POWER. TSCC, SHIELD/CRYO COOLER COMPRESSOR CABINET, & MAGNET MONITOR EQUIPMENT AUTOMATICALLY RESTART AFTER 3 SEC TIME DELAY UPON RESTORATION OF POWER. EMERGENCY OFF LOCKS OUT ALL CONTACTORS.
- IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT USED THEN NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE POWER CABINET
- SUPERVISORY CIRCUIT FOR HVAC INTERLOCK CONTACTS OPEN ON LOSS OF DC POWER OR EMERGENCY OFF OPERATION.



PROTECTIVE DISCONNECT SET–UP
ILLUSTRATION 5–1

5–3–2 System Power Distribution Unit

The PDU Module in the lower portion of ACGD/PDU Cabinet has an integrated filter for a level of power conditioning. The largest allowable phase conductor the PDU will accept is 2/0 AWG (83 mm^2). Larger feeder wires can be connected to the MDP with 2/0 AWG (83 mm^2) between the MDP and PDU.

Note

The ground conductor shall be minimum size of 1/0 AWG copper or the same size as the feeder wire, which ever is larger. Lug connector for the ground wire is to be provided by the contractor, recommended Amp Inc. number 36919 lug.

Note

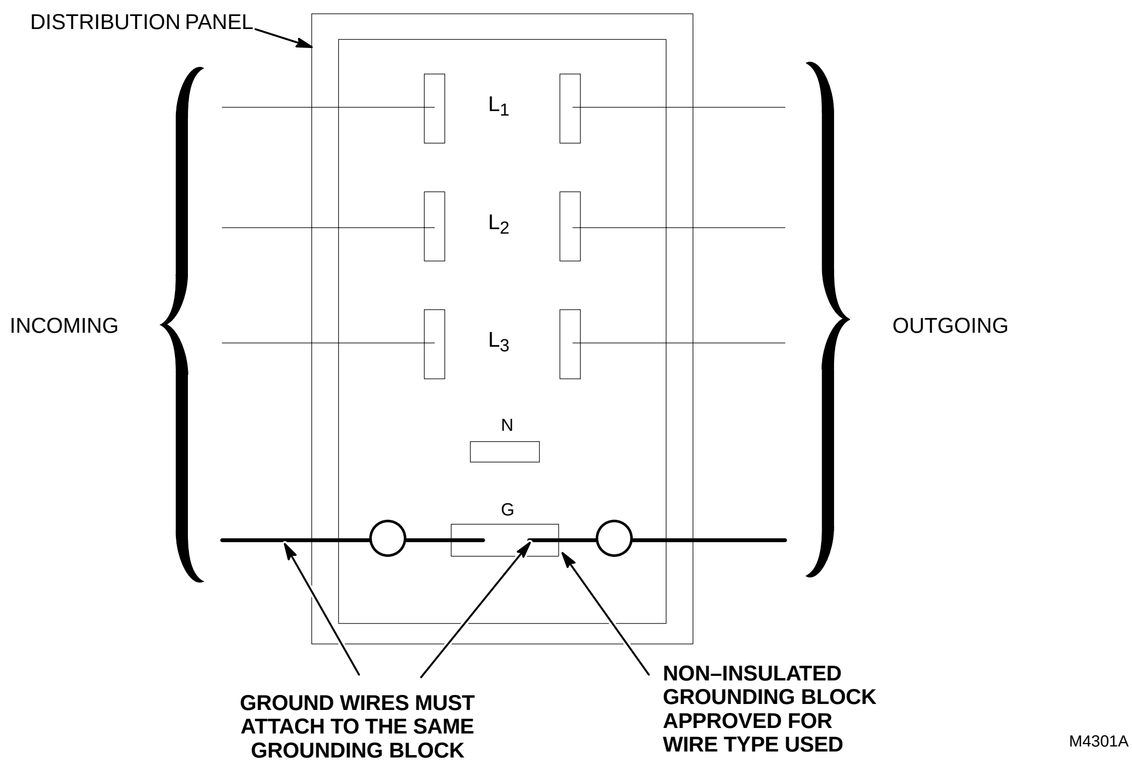
Neutral, if present, must be terminated prior to or inside the Main Disconnect Panel and not brought to the PDU Module in the lower portion of ACGD/PDU Cabinet (MR3).

5–4 GROUNDING

5–4–1 Facility Ground

The ground for the MR system shall originate at the system power source, ie. transformer or first access point of power into a facility, and be continuous to the MR system power disconnect in the room. This ground can be spliced with “High Compression Fittings” and should be terminated at each distribution panel it passes through. When it is broken for a connection to a panel, it shall be connected into an approved non–insulated grounding block with the incoming and outgoing ground in this same grounding block, which is then connected to the steel panel, never using the steel or other material of the panel as the block. See Illustration 5–2.

The connection at the power source shall be at the grounding point of the “Neutral – Ground” if a “Wye” transformer is used, or typical grounding points of separately derived system. In the case of an external facility, it shall be bonded to the facility ground point at the service entrance.



GROUND CONNECTION AT DISTRIBUTION PANEL
ILLUSTRATION 5–2

Ground Wire

The ground wire shall be copper wire with a minimum of AWG 1/0 or the same size as the power feeders whichever is larger. This means that if there is a primary feeder to a distribution panel of 500 MCM with a secondary feeder to the MR system of AWG 2 wire, the ground to the distribution panel shall be 500 MCM with an AWG 1/0 to the MR system. The ground wire impedance from the MR system disconnect, including the ground rod, shall not have an impedance greater than 2 ohms to earth as measured by one of the applicable techniques described in Section 4 of ANSI/IEEE Standard 142 – 1982 which can be accomplished using 3–point Fall Of Potential (3 point measurement) method or Clamp–On Ground Resistance measurement which requires a ground measurement device such as AEMC 3730.

5–4–2 System Ground

The MR system is designed with minimum ground loops to prevent noise currents and natural disturbances from flowing through the low–level signal reference path.

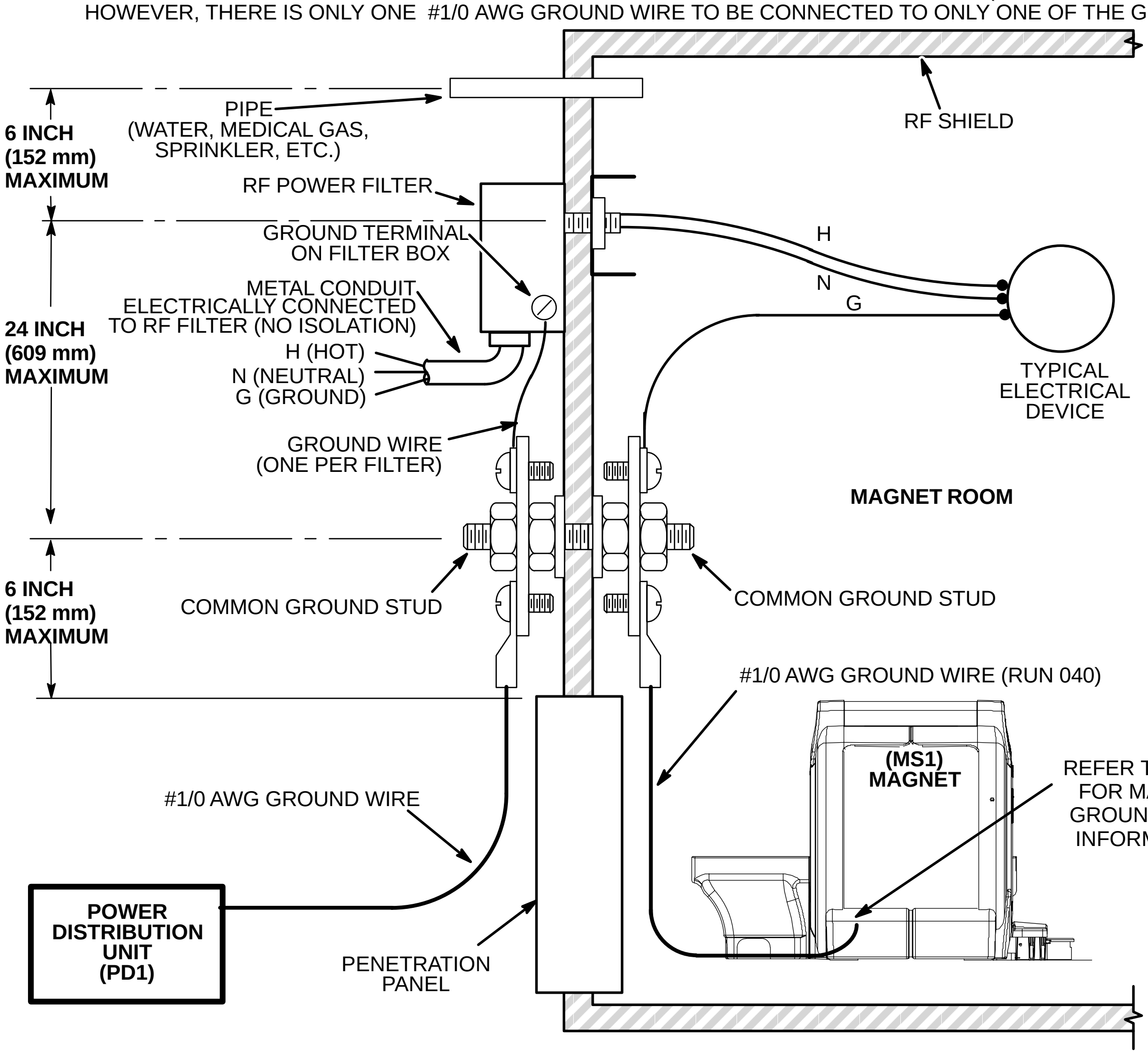
The three major grounding points in the MR system are: the system ground point (bus) in the System PDU (PD1), the enclosure ground points (ground studs located in each cabinet or enclosure), and the RF shielded room common ground point. **This RF shielded room common ground point is to be located within 6 in. (152 mm) of the GE supplied Penetration Panel.** Refer to Section 7, RF SHIELDED ROOM, for a further description of the RF shielded room common ground point.

To ensure patient safety and system performance, the conditions defined in Illustration 5–3 must be met when running power lines into the Magnet Room.

Any modifications or non–MR equipment grounds added to the MR ground system must be approved by your GE Service Representative in order to ensure safety and performance.

5–4–2 System Ground (Continued)

NOTE:
- ALL ITEMS SHOWN ARE CUSTOMER SUPPLIED EXCEPT POWER DISTRIBUTION UNIT, MAGNET, AND TWO #1/0 AWG GROUND WIRES BETWEEN MAGNET GROUND STUD AND RF COMMON GROUND POINT.
- **RESISTANCE BETWEEN ANY TWO GROUNDED DEVICES *MUST NOT EXCEED 0.1 OHM* TO ENSURE EQUAL POTENTIAL GROUND SYSTEM WITHIN MAGNET ROOM.**
- **LOCATE FILTERS WITHIN 2 FEET (600 mm) OF RF COMMON GROUND STUD WHICH MUST BE LOCATED WITHIN 6 INCHES (152 mm) OF PENETRATION PANEL.**
- ALL EXTERNAL CONDUIT MUST BE **METAL AND ELECTRICALLY CONNECTED** TO THE RF POWER FILTERS (IE. NO ISOLATION) UNLESS THE FILTERS ARE LOW VOLTAGE (<30 VOLTS).
- RF POWER FILTERS OF 30 VOLTS OR LESS MAY BE LOCATED ANYWHERE ON THE RF SHIELD **PROVIDED THE INCOMING CONDUIT IS NON–METALLIC OR IS DIELECTRICALLY ISOLATED AND WITH NO GROUND WIRE. IF THE INCOMING CONDUIT IS METALLIC AND/OR HAVE A GROUND WIRE, THESE FILTERS MUST ALSO BE LOCATED WITHIN 24 INCHES (609 mm) OF THE RF COMMON GROUND STUD.**
- ALL CONDUITS IN THE RF ROOM MUST BE **METAL**. STEEL IS ACCEPTABLE PROVIDED IT IS ADEQUATELY ANCHORED.
- ALL ELECTRICAL DEVICES (IE. OUTLETS, LIGHT FIXTURES, ETC.) MUST HAVE A GROUND WIRE FROM ITS POWER SOURCE AND BE GROUNDED TO RF ROOM SHIELD AT THE RF COMMON GROUND STUD AS SHOWN BELOW.
- ALL METALLIC PIPES ENTERING THE RF ROOM, EXCLUDING CRYOGENIC VENT AND FLOOR DRAINS, MUST BE LOCATED WITHIN 30 INCHES (762 MM) OF THE RF COMMON GROUND.
- LCC MAGNET HAS 2 GROUND STUDS, ONE ON FRONT FOOT AND ONE ON REAR FOOT (COLDHEAD SIDE OF MAGNET). HOWEVER, THERE IS ONLY ONE #1/0 AWG GROUND WIRE TO BE CONNECTED TO ONLY ONE OF THE GROUND STUDS.



MR MAGNET ROOM GROUNDING REQUIREMENTS AND TYPICAL DIAGRAM
ILLUSTRATION 5–3

5–5 GROUND FAULT PROTECTION

MR suites and radiology departments are considered health care facilities pursuant to National Electric Code (NEC) 2002 Article 517.2 definitions or NEC1999 Article 517–3 definitions and as such must be powered from sources that comply with the ground fault requirements of NEC 2002 Article 517.17 or NEC 1999 Article 517–17. NEC 2002 Article 517.17(A) or NEC 1999 Article 517–17(a) states “Where ground fault is required for the operation of the service disconnecting means or feeder disconnecting means as specified in NEC 230.95 or 215.10, an additional step of ground fault protection shall be provided in the next level of feeder disconnecting means downstream towards the load.”

NEC 2002 Article 230.95 or 215.10 or NEC 1999 Article 230–95 or 215–10 requires ground fault protection on service disconnecting means rated 1000 Amps or more on solidly grounded WYE services over 150 volts to ground but not over 600 volts phase to phase.

The two or more levels of ground fault shall be coordinated to provide selectivity between each level of ground fault such that a ground fault on the load side of the feeder would cause the feeder and not the service disconnect to open on a ground fault. Six cycles of separation between the different levels of ground fault tripping is required for the system to be considered selective in accordance with NEC 2002 Article 517.17(B) or NEC 1999 Article 517–17(b).

Check national and local electrical codes.